Dmytro Barannik, PhD

Senior / Lead Android Developer

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PROFESSIONAL SUMMARY

Senior / Lead Android Developer & Systems Engineer with **9+ years of commercial expertise** architecting and shipping high-performance mobile applications across video streaming, enterprise IoT, healthcare, and indie commercial software. Deep technical mastery of **Kotlin, Jetpack Compose, Coroutines & Flow, Clean Architecture, and ExoPlayer** streaming ecosystems. Proven track record of engineering for scale: optimized client rendering for millions of active users at **MEGOGO**, led mobile architectures at **CHI Software**, and bootstrapped top-ranked apps with **25,000+ downloads and a 5.0★ rating** as an independent publisher (**DiMALiNA**). Experienced with Android NDK/JNI (C++) and hardware interfaces (BLE, Wi-Fi Direct).

CORE TECH STACK: **Kotlin (Expert), Java**, Jetpack Compose, Material 3, Android SDK & NDK (C++ JNI), Coroutines & Flow, Clean Architecture, MVVM/MVI, ExoPlayer, Room, BLE & Wi-Fi Direct, AI-Augmented Engineering (Autonomous Agent Swarms), Dagger 2 / Hilt.

COMMERCIAL EXPERIENCE

Founder & Lead Mobile Engineer / Publisher — DiMALiNA Studio — Indie Products — 09.2017 – Present

- Bootstrapped and published indie Android products with **25,000+ total downloads** and a **flawless 5.0★ rating** on Google Play (*Tic-Tac-Toe: 3 Player Game*, *Brainrot Clicker*, *People Notes CRM*).
- Architected responsive, modern UIs in 100% **Jetpack Compose and Material 3** with custom canvas physics animations, tactile haptic feedback, and procedural audio.
- Engineered real-time cloud data backends using **Firebase Firestore, Cloud Functions**, and offline-first Room synchronization.
- Integrated full commercial monetization pipelines: **Google Play Billing (In-App Purchases)** and optimized AdMob mediation waterfalls.
- Directed **autonomous AI agent swarms** for accelerated feature delivery, automated UI regression testing, and rapid spec synthesis, tripling production velocity.

Android Developer (High-Load Streaming) — MEGOGO — OTT Streaming — 03.2025 – 11.2025

- Developed and scaled the native Android client application for an international OTT video streaming platform with **millions of active users** across Mobile and Android TV.
- Migrated complex legacy XML screens to declarative **Jetpack Compose**, improving screen rendering speed by **~35%** and eliminating layout stutter during catalog browsing.
- Engineered video playback orchestration using **ExoPlayer**, incorporating low-latency adaptive bitrate streaming (HLS/DASH), channel availability, and parental control locks.
- Maintained **99.8% crash-free sessions** across millions of active streams through rigorous lifecycle scoping, profiler memory leak audits, and automated regression testing.
- Implemented robust deep-linking mechanics via Install Referrer and custom URI schemes to support marketing campaigns and automated user onboarding.

Lead Android Developer / Android Developer — CHI Software — Enterprise Consultancy — 09.2020 – 03.2025

- Led mobile development teams in designing and maintaining complex Android applications in Healthcare, Smart Home, and Security domains using MVVM/MVI and Clean Architecture.
- Built an automated car plate recognition system (ALPR) using Camera 2 API, integrating a native C++ OCR engine via **Android NDK / JNI** with offline Room caching.
- Architected **LiftApp**, an industrial remote hardware configuration tool communicating over **Wi-Fi Direct and BLE** for zero-configuration discovery in noisy RF environments.
- Engineered a full-featured music streaming app with offline background playback, custom notification controllers, and Android Auto integration using ExoPlayer and Room.
- Conducted code reviews, defined coding standards, led technical hiring interviews, and mentored junior and mid-level developers in reactive Kotlin patterns.

Android & Flutter Developer — Nitrix Studio — Commercial Agency — 08.2018 – 08.2020

- Developed cross-platform applications using Flutter (BLoC state management) and native Android apps (Kotlin/Java, MVP/MVVM) across fintech and logistics domains.
- Engineered real-time bidirectional communication using WebSockets and integrated RESTful APIs with OkHttp, Retrofit, and offline SQLite/Room caching.

Android Developer — IT Company — Commercial — 09.2017 – 06.2018

- Developed native Android applications using Java and Kotlin according to Google Material Design guidelines and component lifecycles.
- Managed background services, location tracking, push notifications, and integrated third-party RESTful APIs.

TECHNICAL COMPETENCIES & SKILLS MATRIX

LANGUAGES & CORE
Kotlin (Expert), Java (Advanced), C++ (NDK/JNI), Dart, Python, SQL, Bash scripting.

ANDROID UI & JETPACK
Jetpack Compose, Material 3, Android TV (Leanback), Custom Views, Fragments, ViewBinding, Navigation Component.

ARCHITECTURE & REACTIVE
Clean Architecture, MVVM, MVI, BLoC, SOLID, Kotlin Coroutines, StateFlow, SharedFlow, RxJava 2/3, Multi-Module Setup.

MEDIA & STREAMING
ExoPlayer, VLC, Adaptive Bitrate HLS/DASH, Audio Focus, Background Playback, Android Auto media sessions.

HARDWARE, BLE & NDK
BLE (Bluetooth Low Energy), Wi-Fi Direct, CameraX / Camera 2 API, SerialPort FT232, Android NDK, C++ JNI bridge.

STORAGE & NETWORKING
Room, SQLite, Proto DataStore, Firebase Firestore & Realtime DB, Retrofit 2, OkHttp, WebSockets, WorkManager.

TESTING & PROFILING
JUnit 5, Mockk, Mockito, Espresso, Android Studio Profiler (Memory/CPU/Energy), LeakCanary, Timber logging.

DI, PUBLISHING & OPS
Dagger 2, Hilt, Koin, Google Play Console, Play Billing (IAP), AdMob Mediation, Firebase Crashlytics, Git Flow, CI/CD.

SELECTED MOBILE & SYSTEMS PROJECTS

Tic-Tac-Toe: 3 Player Game ↗ | Kotlin, Jetpack Compose, Material 3, Firebase Firestore, Hilt, Google Play Games
Innovative 3-player multiplayer board game with **10,000+ downloads and a 5.0★ rating**; engineered custom 3-player winning/blocking algorithms, 5 AI difficulty tiers, ranked online matchmaking, and real-time Firestore synchronization.

People Notes: AI Agent CRM ↗ | Kotlin, Jetpack Compose, Coroutines & Flow, Room, Firebase Firestore, WorkManager, Hilt
A privacy-focused offline-first contact and relationship journal with a **5.0★ rating**; features WorkManager background event reminders, multi-tag filtering, SQLite full-text search, and Play Billing integration.

Car Plate Recognition (ALPR) | Kotlin, Android NDK, C++, Camera 2 API, JNI, Room
Real-time automated license plate recognition engine; bridged high-performance native C++ OCR via JNI to analyze high-speed live Camera 2 video frames with zero memory leaks and offline Room detection caching.

LiftApp (Elevator Dashboard & Remote Config) | Java / Kotlin, Android NDK, C++, Wi-Fi Direct, SerialPort FT232, Clean Architecture
Industrial hardware configuration utility and passenger information dashboard; engineered local Wi-Fi Direct parameter synchronization and FT232 SerialPort communication for continuous 24/7 tablet operation.

Lucky Loot Boxes ↗ | Kotlin, Jetpack Compose, Material 3, Clean Architecture, Coroutines, Google Play Billing
Drop simulator game featuring custom Compose unboxing animations, balanced rarity economics, auto-roll progression, haptic feedback, and Google Play Billing (IAP) integration.

EDUCATION & ACADEMIC BACKGROUND

Doctor of Philosophy (PhD) in Information Security · Data Protection & Applied Cryptography — 2021 – 2025
Kharkiv National University of Internal Affairs (KhNUIA)

State Patent UA No. 144599: Positional Video Compression · Co-Inventor — 2020
Ukrainian State Patent Office (Video stream reduction, SIMD AVX2 acceleration)

Master of Science (MS) in Cybersecurity (With Distinction) · Information Security Management — 2019 – 2021
Kharkiv National University of Internal Affairs (KhNUIA)

Bachelor of Science (BS) in Cybersecurity · Information and Communication Systems Security — 2015 – 2019
Kharkiv National University of Internal Affairs (KhNUIA)

LANGUAGES & PROFESSIONAL STRENGTHS

Languages: **English** (Professional Working Proficiency / C1 · Full Technical Communication), **Ukrainian** (Native).
Engineering Leadership: Technical Architecture Design, Code Reviews, Technical Hiring Interviews, Junior & Mid Developer Mentorship, Cross-Functional Team Leadership.